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Advancing Subsea Infrastructure with Non-Metallic Support Structures: Optimizing Design, Logistics, and Sustainability in Shallow Water Installations

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Abstract

This paper explores the application of CSUB's Non-Metallic Subsea Pipeline/Cable Support Structures. All structures are designed using Glass Fiber Reinforced Polymer (GRP) materials. The GRP Support Structures are used mainly for crossings and support of new Pipelines and Cables for Subsea Oil & Gas Developments.

It focuses on design optimization for logistical efficiency through stackability, vessel cost reduction via lightweight materials, subsea stabilization, and innovative post-installation height adjustment using wedges. The study also evaluates the use of GRP Bridges for pipeline crossings to provide more robust crossings and reduce number of installation lifts. Analysis of CO₂ savings using non-metallic material compared to traditional materials will also be addressed.

Introduction

The seabed in the Middle East area has a significant number of existing pipelines. When new offshore fields are introduced, crossing of existing pipelines is a major part of the project development. Old pipelines need to be identified through thorough mapping of the seabed. Pipeline supports, normally steel and concrete sleepers are designed to elevate the new pipeline over the existing pipeline. The size of offshore development projects in the Gulf Area often result in several thousands of sleepers needed to support new pipelines.

The design of the crossings – single or multiple is challenging where main parameters are detailed engineering is to determine each crossing, many of them multiple crossings with main parameters as height of crossing supports, distance between crossings and assessment of seabed bathymetry. The installation phase is challenging as large no of sleepers in various sized needs to be fabricated, transported in several batches to offshore location, and installed one-by-one with high degree of accuracy. Very often, the seabed is not as expected and this will change height of the crossings, To obtain the required pipeline curve, all sleepers need to be placed correctly, if not, the pipeline might not be in contact with one or more of the sleepers, This result in too long and unacceptable freespan. This is normally detected during surveys after pipeline

installation. A new repair campaign is needed to lift and shim sleepers not in contact with the pipeline. Transportation, installation and possible rectification is a very time consuming and expensive operation.

Non-Metallic/GRP Sleepers can be made stackable and this reduce transportation time, increase efficiency due to reduced deck space requirement and by that save considerable cost and CO₂. For GRP Sleepers, CSUB has developed wedges that can be retrofitted on sleepers not in contact with the pipeline. This will simplify the mitigation time,

We will in this paper also address the possibility of replacing a high number of sleepers with fewer pipeline crossing bridges. Typically, 4 sleepers in a single line crossing can be replace with one bridge, For multiple crossing, a solution with ramp and straight bridges can simplify the crossing. This will result in a drastically reduction of vessel days for the installation phase. Another advantage is that a bridge is not so vulnerable for uneven seabed as single sleepers and risk for later mitigation work is reduced.

Objectives and Scope

This paper explores the application of CSUB's Non-Metallic Subsea Pipeline/Cable Support Structures. All structures are designed using Glass Fiber Reinforced Polymer (GRP) materials. The GRP Supports Structures are used mainly for crossings and support new Pipelines and Cables for Subsea Oil & Gas Developments.

It focuses on design optimization for logistical efficiency through stackability, vessel cost reduction via lightweight materials, subsea stabilization, and innovative post-installation height adjustment using wedges. The study also evaluates the use of GRP Bridges for pipeline crossings to provide more robust crossings and reduce number of installation lifts. Analysis of CO₂ savings using non-metallic material compared to traditional materials will also be addressed.

Literature Review

Non-Metallic materials as Glass Fiber Reinforced Polyester (GRP) is increasingly being explored in various engineering applications due to its favorable properties, such as high strength-to-weight ratio, corrosion resistance, and durability. Studies have shown that GRP structures can withstand harsh subsea environments, making them suitable for use in pipeline supports and other subsea applications.

- **Material Properties and Performance:** Research indicates that GRP can offer substantial weight savings compared to steel, which can reduce installation costs and risks associated with heavy lifting operations. Additionally, GRP's corrosion resistance ensures a longer service life and reduced maintenance requirements.
- **Applications in Subsea Engineering:** Several studies have demonstrated the effectiveness of GRP in subsea structures. For instance, GRP has been successfully used in the construction of subsea protection structures, riser guards, crossing bridges, and other components, showcasing its versatility and reliability in challenging conditions.
- **Environmental Impact:** The use of GRP contributes to lower CO₂ emissions during the production and installation phases compared to traditional steel solutions. This aligns with the industry's growing emphasis on sustainability and reducing the environmental footprint of oil and gas operations.

Methods, Procedures, and Process

CSUB has maintained a strong focus on research and development (R&D) of GRP (Glass Fiber Reinforced Polyester) products since its inception. In 2016, the company strategically shifted towards developing GRP alternatives to traditional steel solutions, establishing an internal engineering department that now comprises

With a combination of single and multiple crossings, we found several projects with over 1000 sleepers in up to 100 variants.

Focus area 1 – Stackability and reduction of requirement for storage and deck space using Non-Metallic Supports

Non-Metallic/GRP Sleepers

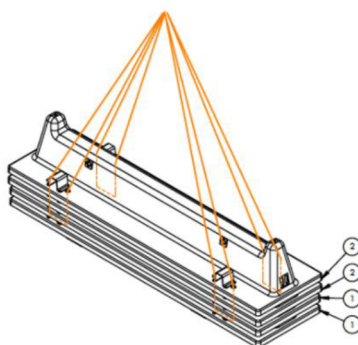


Figure 3—Stacking of GRP sleepers

Non-Metallic sleepers have its strength in the shell, ie it is hollow. This enable stacking of the sleepers as shown above. This will drastically reduce the area requirement for storage and offshore transportation.

Non-Metallic sleepers have integrated spacers to ease stacking and avoid removal of stacking materials subsea. Stacks can also be deployed directly to wet storage, and installation campaign of each sleepers can be performed subsea saving valuable vessel time.

CSUB GRP Sleepers can also be designed to allow for retrofit of our patented wedge-system. This is developed to adjust the height of the sleepers if contact is not achieved after installation of pipeline.

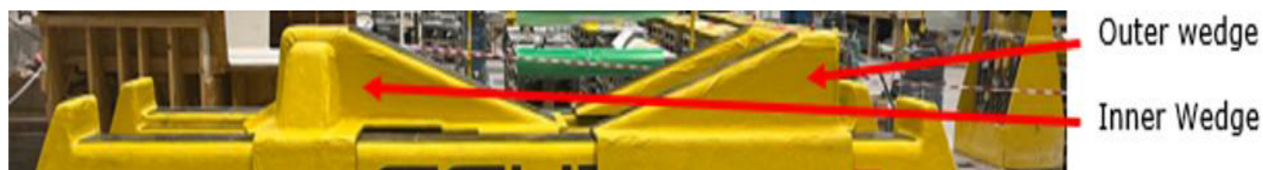


Figure 4—Wedge design to allow height adjustment post installation

This will reduce time for adjustment of height of the sleepers after installation survey and avoid lifting the pipeline and install shims to adjust the height of the sleeper.

‘Non-Metallic - GRP Bridges for Cable and Pipeline Crossings

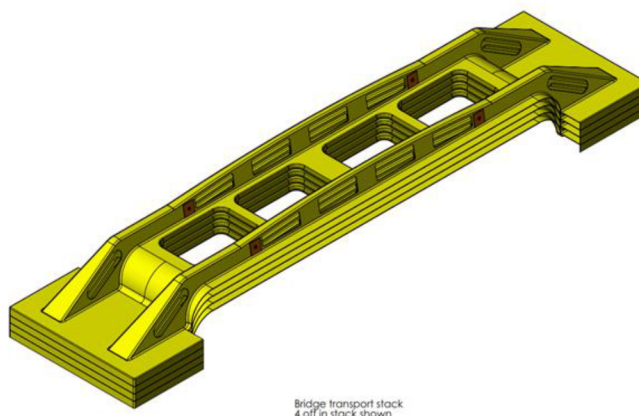


Figure 5—Stacking of GRP Bridges

As for sleepers, Non-Metallic bridges have its strength in the shell, and is stackable as shown above. This will drastically reduce the area requirement for storage and offshore transportation.

Focus Area 2 replacing high number of sleepers with bridges

With individual sleepers, distance between each unit, individual height and seabed bathymetry is vital to obtain correct pipeline routing. Using Non-Metallic bridges for pipeline crossing will highly reduce no of sleepers and also be more robust for uneven seabed.

Example of single crossing solution below with individual sleepers vs one single bridge

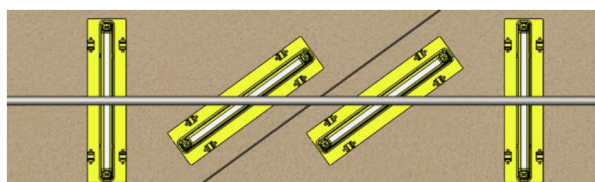


Figure 6—Crossing with sleepers

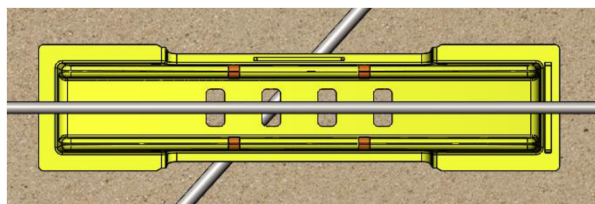


Figure 7—Same crossing with one bridge instead of 4 sleepers

Cost for single bridge is higher compared to individual sleepers, however with only one lifting operation and more robustness to uneven seabed, the solution can be very competitive.

For multiple crossing, the reduction of no of units is considerably higher.

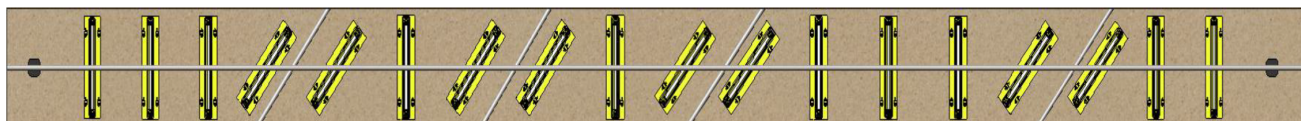


Figure 8—Crossing with sleeper solution

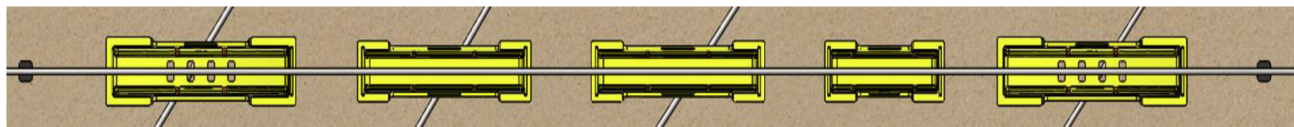


Figure 9—Same crossing with bridges instead of sleepers

Focus area 3 stabilization of the Non-Metallic sleepers and bridges

As structures in Non-Metallic / GRP material normally have considerably lower weight compared to steel/concrete sleepers, determination of ballast weight needed to ensure stability is essential.

CFD analysis was performed to determine the Drag force, Lifting Force and overturning moment of the sleepers and GRP Bridges. Optimization with shape and perforation was performed to reduce the drag/lift/overturning moment of the Non-Metallic structures.

Results of this CFD were used to develop a spreadsheet model to be able to calculate required ballast weight to ensure stability for a variety of Sleepers and Bridges.

Results

Focus area 1 – Stackability and reduction of requirement for storage and deck space using Non-Metallic Supports

Sleepers. Non-Metallic/GRP sleepers can not compete with concrete sleepers for pure purchasing cost, however if stackability/installation and transportation cost / time is taken into consideration, the solution can be competitive. For extremely shallow water (below 15-20 m) concrete sleepers might have an advantage due to weight for stabilization.

Non-Metallic/GRP Sleepers can be competitive with steel sleepers, especially if stackability/installation and transportation cost and time is taken into consideration.

Estimation of area savings evaluation based on 8 sleepers to be stacked compared to traditional non-stackable solutions. Dimensions of sleepers can vary typically from 4 – 50 m of length. Size of projects in GCC are normally quite high – from 100-1000 sleepers is quite normal. For this estimate we have used 400 sleepers 15 × 4 m sleeper giving an area of 60 m² pr sleeper/stack. With 400 sleepers the total area will be 24000 m², with 8 pr stack, this will be 3000 m².

Storage on shore: Considerable saving of more than **20000 m²** valuable land. It will also enable quicker loadout as more sleepers can be made available at quay side prior to vessel/barge arrival,

Vessel Utilization – If we estimate a deck area for barge/vessel of 1500 m² and a sleeper/stack footprint of 60 m² and adding 20% for extra space between each sleeper/stack, approx. 20 sleepers/stacks can be transported pr trip. For Non-stackable sleepers approx. 20 transports are required (400ea/20). For stackable Non-Metallic/GRP solution, 160 sleepers can be transported pr trip, ie 3 transports compared to 20 with individual sleepers. This if cause a simplified estimate as other areas like location of sleepers etc probably will result in more trips, however a considerable saving is possible to cost/time and CO₂ footprint.

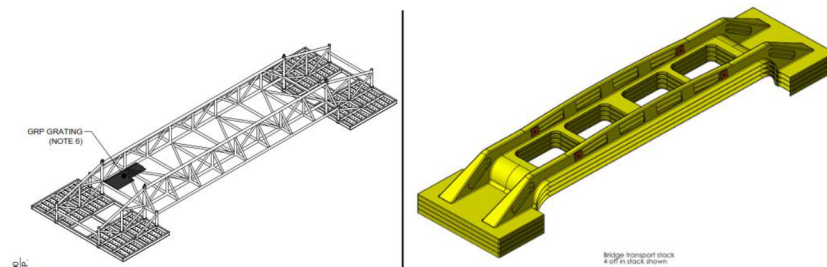
Non-Metallic - GRP Bridges

For Non-Metallic / GRP bridges, the cost for purchasing can be competitive with steel bridges, especially with higher quantities. Use of moulds and considerably shorter production time will also be highly

advantageous to the project execution schedule. The Non-Metallic GRP Bridges are normally installed upended increasing the allowable seastate for installation.



Typical example of area savings for a steels crossing bridge:



Example above shows a 45×10 m bridge – footprint 450 m².

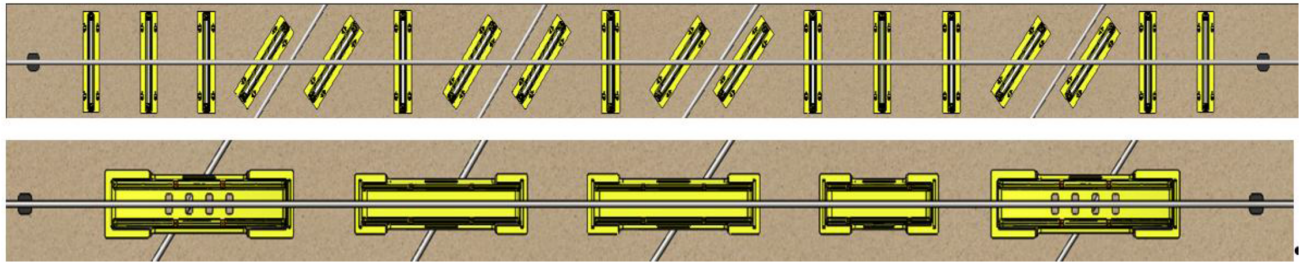
A project for crossing bridges in GCC is typically +/-60 bridges pr project and we assume 5 bridges pr stack.

Storage onshore: For all 60 bridges, 450×60 bridges gives a total requirement of 27.000 m² net for onshore storage. For a stackable solution with 5 bridges pr stack, the requirement will be 4.500 m², giving a net saving in excess of **20000 m²** valuable storage area. It will also enable quicker loadout as more bridges can be made available at quay side prior to vessel/barge arrival,

Vessel Utilization, With a barge / vessel of 1500 m², 3 bridges can be transported pr trip, ie 20 trips for Non – Stackable bridges. For stackable Non-Metallic / GRP bridges, we can assume 3 stacks pr vessel if crane is rigged with auxiliary winch to upend the bridge directly from stack. With a stackable solution, a reduction in trips from 20 down to 5 is possible. This if cause a simplified estimate as other areas like location of Non-Metallic / GRP Bridges probably will result in a few more trips, however a considerable saving is possible to cost/time and CO₂ footprint.

Focus area 2 – Replacing sleepers with bridges

We performed a study for a concrete project with more than 600 crossings An estimated number of sleepers for single and multiple crossings were estimated to approx. 3.500 sleepers. If Non-Metallic GRP Bridges are used with principles as shown below, the estimated number of bridges is approx. 200.



Information provided to CSUB is that an average of 4 sleepers can be assumed installed pr vesselday. Using this, we will estimate 890 vesseldays to install all sleepers. As the sleepers need high degree of accuracy for installation, we have assumed that we will have same no of GRP Bridges installed pr vessel day, ie a total of 200 vesseldays needed. The purchasing cost for Non-Metallic GRP Bridges are higher than sleepers, however with the considerable saving in vesseldays, the overall result will provide significant reduction of cost and CO2 emissions.

Focus area 3 stabilization of the Non-Metallic sleepers and bridges

Detailed analysis using CFD methods and calculations were performed to determine the loads from subsea current and wave induced current. Worst case scenario with environmental forces 90 degrees to the direction of the structure was chosen for all analysis

For Non-Metallic GRP Sleepers and Bridges, full body of the structures were used as basis, followed by analysis of effect of cut-outs the structures to reduce the drag and lift caused by subsea current.

Sleepers.

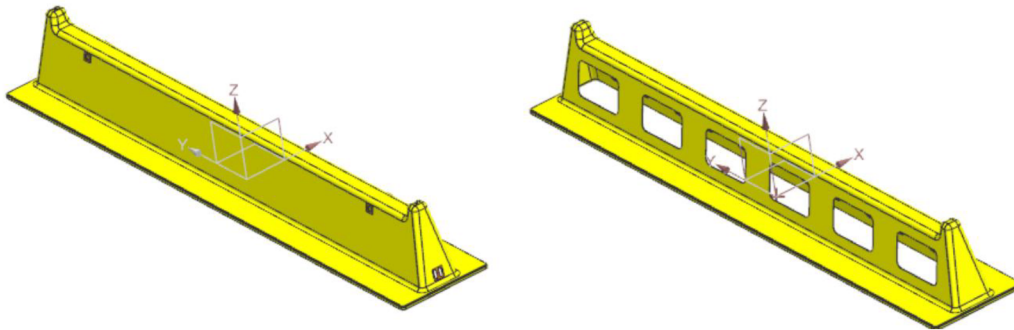


Figure 10—Sleeper without and with cutouts to reduce environmental loads

Environmental data used in the analysis:

- Water depth 10 m
- Wave Height 7.3 m
- Wave Period 8.6 s
- Current Velocity 0.69 m/s
- Wave Induced Current 2.96 m/s (calculated)

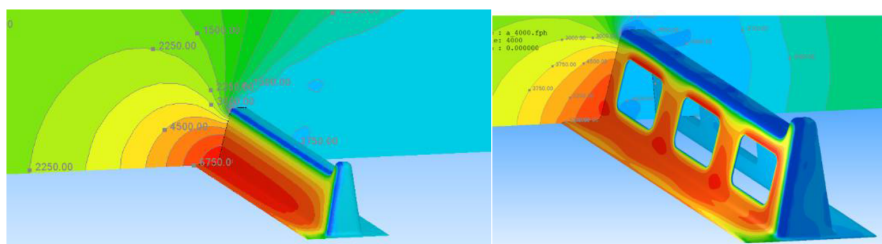


Figure 11—CFD results

The drag coefficient of the original sleeper geometry without cut-outs was around 1.32. With cut-outs, the drag coefficient was consistently around 0.89.

Both geometries are subject to downward vertical force. This is due to high pressure areas on the top of the mud plate originating from stagnation against the upstream wall of the sleeper body. This counteracts the upward suction on top of the sleeper.

Simplified sleeper with cut outs , Vc 3.65m/s, No Free surface, grid2	241730	-23757	0.890	-0.192	4726	-22737	237010	-1020
Simplified sleeper with cut outs , Vc 3.65 m/s, No Free surface, grid2, unsteady	241110	-28556	0.887	-0.231	4760	-20089	236350	-8467
Simplified original sleeper , Vc 3.65m/s, No Free surface, grid2	316060	-756	1.163	-0.006	4844	-50674	311220	49918

Summary of observations:

Drag (FXtotal) reduced from approx. 316kN to 241kN, ie a reduction of 23%

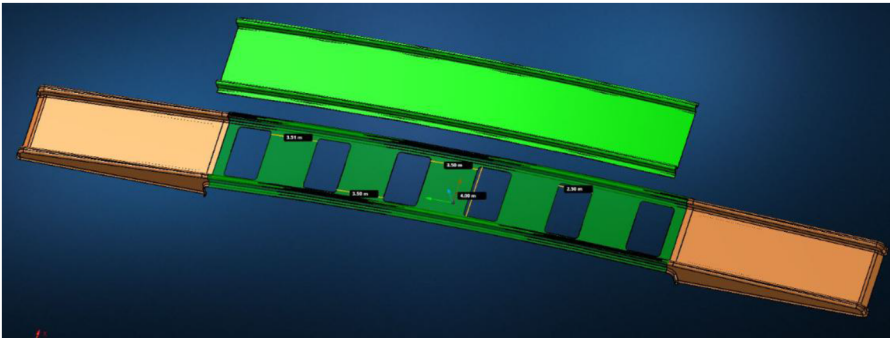
Lift force (FZSIBody) reduced from approx. 50kN to slightly become negative, ie downforce.

Above analysis shows that by use of change in design using cut-outs, the drag/lift force can be considerably reduced giving a substantial reduction in ballast needed to stabilize the sleeper.

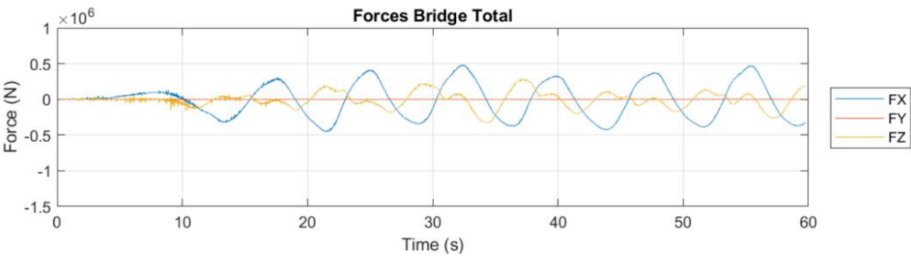
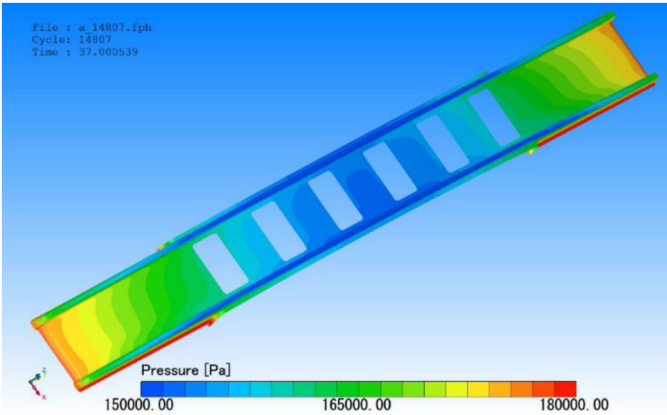
CSUB also checked the effect of 3 sleepers. This shows a significant interaction effect with reduced drag. For the mid and aft sleepers, the drag is reduced by 80%, and even the first sleeper shows a reduction of about 11%.

	Mid sleeper				Forward sleeper		Aft sleeper	
	FX_X0	FZ_X0	Cd_X0	Cl_X0	FX_Xmin20	FZ_Xmin20	FX_X20	FZ_X20
	(N)	(N)	(-)	(-)	(N)	(N)	(N)	(N)
Simplified sleeper with cut outs , Vc 3.65m/s, No Free surface, grid2 X 3	49227	29484	0.181	0.239	214750	-42982	45772	35117

Bridges. A typical Non-Metallic GRP Bridge was analysed to determine the effect of introduction of large cut-outs on the center part of the arch of the bridge. It was not expected significant effect for drag forces, however lift force should be reduced.



Typical environmental data used was:
Water depth 11 and 20 m
Wave Height 7.3 and 4.7 m
Wave Period 8.6 and 7.9 m
Pressure distribution



Bridges		Type 2						
Report Section		4.1.1		4.1.2		4.2		
Bridge Arch		Cut Out	Cut Out	Cut Out	Cut Out	Full	Full	Full
Water Depth	m	11.3	11.3	20	20	11.3	11.3	20
WaveHeight Nominal Max	m	7.3	4.7	5.3	5.3	7.3	4.7	5.3
Wave Period	s	8.6	7.9	7.6	7.6	8.6	7.9	7.6
FXTotal max	N	1.01E+06	8.13E+05	6.93E+05	4.84E+05	1.03E+06	8.31E+05	6.90E+05
FXTotal min	N	-7.11E+05	-6.80E+05	-6.28E+05	-4.53E+05	-8.04E+05	-7.21E+05	-6.34E+05
FZTotal max	N	3.61E+05	3.07E+05	3.48E+05	2.82E+05	7.01E+05	6.28E+05	7.60E+05
FZTotal min	N	-7.50E+05	-5.16E+05	-4.65E+05	-3.26E+05	-1.39E+06	-1.01E+06	-7.43E+05

Drag forces (FXTotal max) remains as expected close to the same with or without cut-outs.

FZTotal max is reduced with approx 50% due to the large cut-outs.

The above cut-outs will be implemented in all CSUB Non-Metallic GRP Bridges to reduce the requirement for stabilization with steel ballast or concrete blocks.

Discussion

This study indicates that non-metallic (GRP) subsea supports can address persistent challenges in Gulf projects where dense legacy corridors drive high crossing counts and tight installation windows. The key differentiators are logistics efficiency from stackability and lower unit weight, operational flexibility via retrofit wedges for post-installation height adjustment, and the option to replace many sleepers with bridges—together targeting major cost drivers such as deck space, lifts, and vessel days.

The logistics gains are material at project scale. For a representative sleeper campaign (~ 400 units, $\sim 15 \times 4$ m), stackable GRP reduces onshore storage from $\sim 24,000$ m² to $\sim 3,000$ m² and can cut offshore transports from ~ 20 trips to ~ 3 (acknowledging layout and sequencing caveats). These reductions compress campaign duration and lower CO₂ from marine spreads.

Bridge-based crossings add another lever by consolidating numerous individually placed sleepers into fewer, more robust units that are less sensitive to seabed unevenness. For typical Gulf bridge programs, stackability similarly reduces storage requirements (e.g., $\sim 27,000$ m² to **$\sim 5,400$ m²** in the example) and can lower vessel trips (e.g., ~ 20 to ~ 4), improving schedule certainty.

A concrete baseline study with >600 crossings estimated $\sim 3,500$ sleepers; an equivalent GRP bridge concept reduced the scope to ~ 200 bridges. Using the industry rule-of-thumb of ~ 4 sleeper installs per vessel day, this corresponds to ~ 890 vessel days for sleepers versus ~ 200 for bridges. Although bridge unit cost is higher, the vessel-day savings dominate the total installed cost and footprint in many scenarios.

Hydrodynamic stability is a legitimate concern for lightweight composites but can be managed through geometry and ballast. CFD shows that adding cut-outs lowers sleeper drag by $\sim 25\%$ and nudges net lift slightly downward, reducing ballast demand. In multi-sleeper lines, downstream units also benefit from shielding, with drag reductions up to $\sim 80\%$. For bridges, large arch cut-outs roughly halve the vertical load (FZ) with little change in drag, further easing ballast requirements. These results support perforations and shape optimization as standard features of GRP designs.

GRP is not a universal replacement. In ultra-shallow waters ($\sim <20$ m), the intrinsic mass of concrete can be advantageous for stability. In addition, broader regional deployment of GRP will benefit from continued local qualification, supply-chain scaling, and stakeholder familiarity. Early engagement with operators, installers, and regulators—along with standardization of GRP sleeper and bridge families (including wedge-ready variants)—will help de-risk deployment and compress schedules.

Future work should prioritize field monitoring of installed GRP crossings to validate CFD-based ballast models under site-specific metocean spectra, and a quantified life-cycle assessment comparing GRP, steel, and concrete across fabrication, transport, installation, and intervention phases. (The spreadsheet model derived from the CFD provides a practical starting point for such validation.)

Conclusion

Non-metallic GRP supports are a viable—and often advantageous—option for shallow-water pipeline and cable crossings in the Gulf, delivering consistent gains in logistics and installation efficiency when stability is engineered with appropriate ballast. Key takeaways are:

- **Logistics efficiency.** Stackable GRP sleepers and bridges sharply reduce storage area and offshore transports, compressing campaign duration and CO₂ footprint.

- **Installation productivity.** Replacing many sleepers with fewer bridges can reduce vessel days by multiples (e.g., $\sim 890 \rightarrow \sim 200$ in the studied case), while improving tolerance to uneven seabeds.
- **Operational flexibility.** Retrofit wedge systems enable rapid, targeted height corrections post-lay without heavy re-lifting campaigns.
- **Engineered stability.** Geometry/perforation optimization demonstrably lowers hydrodynamic loads (e.g., $\sim 25\%$ drag reduction for sleepers; $\sim 50\%$ vertical-load reduction for bridges), reducing ballast requirements.

Concrete may remain preferable in very shallow, high-energy sites due to inherent mass, and broader regional deployment of GRP will benefit from continued qualification and capacity building. Nonetheless, the demonstrated reductions in vessel days, lifts, and deck footprint position GRP sleepers and bridges as strategic tools for accelerating subsea development with lower total installed cost and environmental impact in the Middle East.